

TECHNICAL DATASHEET

CPS-001 - RoboSoldagemAlfa
Cyber-Physical System Technical Datasheet

Prototype technical datasheet generated from the CPS Asset Administration Shell (AAS). It is not an official manufacturer datasheet.

1. CPS Identification

Attribute	Value	AAS source
CPS ID	CPS-001	assetInformation
Asset Name	RoboSoldagemAlfa	assetInformation
Asset Type	Welding Robot 6-Axis	assetInformation
Manufacturer	IndustrialRobotics Corp	assetInformation
Serial Number	SN-WELD-9921	assetInformation
Global Asset ID	urn:asset:cps1:robosoldagem_alfa_v3	assetInformation
AAS ID	urn:uuid:cps1-aas-001	AssetAdministrationShell
AAS idShort	CPS1_AAS	AssetAdministrationShell
Asset Kind	Instance	assetInformation

2. Technical Data

Attribute	Value	AAS source
MaxPayload	10kg	TechnicalData
DegreesOfFreedom	6	TechnicalData
WeldingProcess	Arc Welding	TechnicalData
NominalCycleTimeMs	1800	TechnicalData

3. Functional Capability

Attribute	Value	AAS source
Identifier	execute_welding_cycle	Functions
Name	Execute Welding Cycle	Functions.execute_welding_cycle
Description	Execução do ciclo de soldagem com controle de temperatura da ponta, corrente de arco e pressão de gás.	Functions.execute_welding_cycle
Allowed statuses	active maintenance awaiting_replacement failure	Functions.execute_welding_cycle

4. Communication Interfaces

Attribute	Value	AAS source
PrimaryProtocol	MQTT	AssetInterfacesDescription
BrokerHost	broker.hivemq.com	AssetInterfacesDescription
BrokerPort	1883	AssetInterfacesDescription
BaseTopic	cps1	AssetInterfacesDescription
WebSocketInterfaces.MQTTWS	ws://broker.hivemq.com:8000/mqtt	WebSocketInterfaces
WebSocketInterfaces.MQTTWSS	wss://broker.hivemq.com:8884/mqtt	WebSocketInterfaces
CommandInterface.Topic	cps1/cmd	CommandInterface
CommandInterface.Direction	Inbound	CommandInterface
StatusInterface.Topic	cps1/status	StatusInterface
StatusInterface.Direction	Outbound	StatusInterface
AckInterface.Topic	cps1/ack	AckInterface
HealthInterface.Topic	cps1/health	HealthInterface
DataInterface.Topic	cps1/data	DataInterface
DataInterface.Direction	Outbound	DataInterface
OOEInterface.Topic	cps1/oeo	OOEInterface
OOEInterface.Direction	Outbound	OOEInterface
ACSMOOEInterface.Topic	acsm/cps1/oeo	ACSMOOEInterface
ACSMOOEInterface.Direction	Outbound	ACSMOOEInterface
ACSMOOEInterface.Producer	CPS-1 / Node-RED	ACSMOOEInterface
ACSMOOEInterface.Consumer	ACSM	ACSMOOEInterface
ACSMLearningInterface.Topic	acsm/cps1/learning	ACSMLearningInterface
ACSMLearningInterface.Direction	Outbound	ACSMLearningInterface
ACSMLearningInterface.Producer	CPS-1 / Node-RED	ACSMLearningInterface
ACSMLearningInterface.Consumer	ACSM	ACSMLearningInterface
RESTEndpoints.DescriptionEndpoint	http://localhost:1881/api/cps1/description	RESTEndpoints
RESTEndpoints.SummaryEndpoint	http://localhost:1881/api/cps1/summary	RESTEndpoints
RESTEndpoints.IndicatorsEndpoint	http://localhost:1881/api/cps1/indicators	RESTEndpoints
RESTEndpoints.HistoryEndpoint	http://localhost:1881/api/cps1/history	RESTEndpoints
RESTEndpoints.HealthEndpoint	http://localhost:1881/api/cps1/health	RESTEndpoints

5. Lifecycle Integration

Lifecycle phases are distinct from operational conditions. Plug, Play-Evolution, and Unplug are lifecycle phases; running, stopped, and maintenance are operational conditions and are not treated as lifecycle phases in this document.

Attribute	Value	AAS source
SupportedPhases	plug, play, monitor, unplug	LifecycleIntegration
CurrentPhase	play	LifecycleIntegration

6. ACSM Integration

Attribute	Value	AAS source
ACSMAssetId	ACSM-CPS1-001	ACSMIntegration
NodeRedFlowReference	flow:cps1-welding-logic	ACSMIntegration
DashboardURL	http://localhost:3007/cps1	ACSMIntegration
AnalyticsIntegrationMode	MQTT + AAS	ACSMIntegration
OEEPublishTopic	acsm/cps1/oe	ACSMIntegration
LearningPublishTopic	acsm/cps1/learning	ACSMIntegration
OEESubmodelRef	urn:uuid:submodel:cps1:oeemonitoring	ACSMIntegration
LearningSubmodelRef	urn:uuid:submodel:cps1:learning	ACSMIntegration

7. Digital Representation

Attribute	Value	AAS source
Representation	Asset Administration Shell	AssetAdministrationShell
Global Asset ID	urn:asset:cps1:robosoldagem_alfa_v3	assetInformation
AAS instance ID	urn:uuid:cps1-aas-001	AssetAdministrationShell
Document endpoint	/api/cps1/datasheet/pdf	Documents.DatasheetPDF

8. Document Information

Attribute	Value	AAS source
Document type	Prototype Technical Datasheet	Generated document
Source	CPS-001 Asset Administration Shell	Source definition
Generated for	PPU Prototype	Requested document scope
Date generated	2026-08-30	Generation date

Excluded runtime data

Current OEE, health, temperature, arc current, gas pressure, counters, Learning, Reasoning, Prediction, and Recommendation are runtime data or intelligence outputs. They are deliberately excluded from this static technical specification.